Topic 5 – Energy Flow, Ecosystems and the Environment

Candidates will be assessed on their ability to:

5.1	understand the overall reaction of photosynthesis as requiring energy from light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide and releasing oxygen into the atmosphere
5.2	understand how photophosphorylation of ADP requires energy and that hydrolysis of ATP provides an immediate supply of energy for biological processes
5.3	understand the light-dependent reactions of photosynthesis, including how light energy is trapped by exciting electrons in chlorophyll and the role of these electrons in generating ATP, reducing NADP in cyclic and non-cyclic photophosphorylation and producing oxygen through photolysis of water
5.4	(i) understand the light-independent reactions as reduction of carbon dioxide using the products of the light-dependent reactions (carbon fixation in the Calvin cycle, the role of GP, GALP, RuBP and RUBISCO) (ii) know that the products are simple sugars that are used by plants, animals and other organisms in respiration and the synthesis of new biological molecules (polysaccharides, amino acids, proteins, lipids and nucleic acids)
5.5	understand the structure of chloroplasts in relation to their role in photosynthesis
5.6	understand what is meant by the terms *absorption spectrum* and *action spectrum*
5.7	understand that chloroplast pigments can be separated using chromatography and the pigments identified using Rf values
5.8	**CORE PRACTICAL 10** **Investigate the effects of light intensity, light wavelength, temperature and availability of carbon dioxide on the rate of photosynthesis using a suitable aquatic plant.**
5.9	(i) understand the relationship between gross primary productivity (GPP), net primary productivity (NPP) and plant respiration (R) (ii) be able to calculate net primary productivity
5.10	know how to calculate the efficiency of biomass and energy transfers between trophic levels
5.11	understand what is meant by the terms *population*, *community*, *habitat* and *ecosystem*
5.12	understand that the numbers and distribution of organisms in a habitat are controlled by biotic and abiotic factors
5.13	understand how the concept of niche accounts for the distribution and abundance of organisms in a habitat

5.14	**CORE PRACTICAL 11** **Carry out a study of the ecology of a habitat, such as using quadrats and transects to determine the distribution and abundance of organisms, and measuring abiotic factors appropriate to the habitat.**
5.15	understand the stages of succession from colonisation to the formation of a climax community
5.16	understand the different types of evidence for climate change and its causes, including records of carbon dioxide levels, temperature records, pollen in peat bogs and dendrochronology, recognising correlations and causal relationships
5.17	understand the causes of anthropogenic climate change, including the role of greenhouse gases in the greenhouse effect
5.18	understand how knowledge of the carbon cycle can be applied to methods to reduce atmospheric levels of carbon dioxide
5.19	(i) understand that data can be extrapolated to make predictions and that these are used in models of future climate change (ii) understand that models for climate change have limitations
5.20	understand the effects of climate change (changing rainfall patterns and changes in seasonal cycles) on plants and animals (distribution of species, development and lifecycles)
5.21	understand the effect of temperature on the rate of enzyme activity and its impact on plants, animals and microorganisms, to include Q_{10}
5.22	**CORE PRACTICAL 12** **Investigate the effects of temperature on the development of organisms (such as seedling growth rate or brine shrimp hatch rates), taking into account the ethical use of organisms.**
5.23	understand how evolution (a change in allele frequency) can come about through gene mutation and natural selection
5.24	understand how isolation reduces gene flow between populations, leading to allopatric or sympatric speciation
5.25	understand the way in which scientific conclusions about controversial issues, such as what actions should be taken to reduce climate change, or the degree to which humans are affecting climate change, can sometimes depend on who is reaching the conclusions
5.26	understand how reforestation and the use of sustainable resources, including biofuels, are examples of the effective management of the conflict between human needs and conservation